

A Computational Systems Model of Multi-Level Governance for Human Flourishing under Bounded Non-Exploitation

Harris Wang
Athabasca University, Athabasca, Canada
harrisw@athabascau.ca

Abstract

Governance operates simultaneously across multiple levels—from local communities to national governments to international institutions—yet most formal governance models treat governance as a single-level phenomenon and permit jurisdictions to advance their own flourishing at the expense of others. This conceptual paper extends the General Governance Success Model (GSM) to account for multi-level governance under a **bounded non-exploitation constraint**: no jurisdiction may impose *uncompensated, non-trivial* sacrifices on other jurisdictions for its own benefit. The extended model represents governance at L levels, each characterized by five coupled capabilities (accountability, institutional competence, social cohesion, strategic continuity, and adaptive learning). The framework formalizes vertical and horizontal interactions decomposed into cooperative and exploitative components, cross-level information flows with fidelity matrices, level-specific dynamics, recursive learning, and a formal non-exploitation constraint that admits both **threshold tolerance** and **compensation** as legitimate exceptions. A **formal welfare function** is specified, and the **threshold parameter** τ_j is decomposed into descriptive contextual variables and normative parameters. **Compensation is modelled as a vector** of transfers (fiscal, infrastructural, representational, regulatory, and future-oriented). A **roadmap for operationalizing the exploitation index** is provided. The model generates testable hypotheses concerning vertical alignment, polycentric governance, cross-level coordination, recursive learning, and non-exploitation. An **illustrative worked example** applying the GSM to Denmark—using real raw governance indicators but **explicitly illustrative coefficients**—is presented for pedagogical purposes, with comparative rankings relegated to a brief supplementary illustration. To support replication and application, the model is released as an open-source Python library, `gsm-governance`, which implements the state-space dynamics, bounded non-exploitation constraint, vector-valued compensation, threshold decomposition, and all reported indices. The extended GSM provides a foundation for analysing governance systems in their full institutional complexity while respecting normative constraints on inter-jurisdictional justice.

Keywords: multi-level governance; complex adaptive systems; dynamical systems; polycentric governance; Viable System Model; non-exploitation; distributive justice; human flourishing; bounded rationality

1. Introduction

1.1 The Challenge of Multi-Level Governance

Governance does not occur at a single level. It operates simultaneously across nested scales: local communities, municipal governments, regional authorities, national states, and supranational institutions. Each level has distinct capabilities, constraints, and functions, yet they are deeply interconnected through vertical and horizontal relationships. Policy decisions at one level constrain or enable decisions at others; information flows across levels; legitimacy and accountability are distributed; and crises cascade through the system.

A critical but often neglected feature of this interconnectedness is that **jurisdictions can impose costs on one another**. A higher-level government may extract resources from lower-level jurisdictions; a lower-level jurisdiction may externalize environmental harms to neighbours; a peer jurisdiction may engage in regulatory competition that degrades collective welfare. In each case, one jurisdiction advances its own flourishing at the expense of others—a phenomenon we term **jurisdictional exploitation**.

Despite this multi-level reality, most formal governance models treat governance as a single-level phenomenon. Even frameworks that recognize multi-level structures rarely impose constraints on cross-jurisdictional harm. This simplification obscures the complexity of real governance systems and permits analytical frameworks that would endorse Pareto-improving exploitation—outcomes in which some jurisdictions gain by making others worse off.

1.2 The Computational and Systems Perspective

This paper extends the General Governance Success Model (GSM) to account for multiple levels of governance, subject to a **bounded non-exploitation constraint**: no jurisdiction may impose uncompensated, non-trivial sacrifices on other jurisdictions for its own benefit.

The GSM is a systems-theoretical metamodel that represents governance as a complex adaptive and information-processing system composed of five interacting institutional capabilities—accountability, competence, social cohesion, strategic continuity, and adaptive learning—whose dynamic configuration and contextual alignment determine the capacity of a governance system to promote sustainable human flourishing (Wang, 2026a).

The extension draws on six intellectual traditions:

1. **The Viable System Model (VSM)**: Beer's (1972, 1979) cybernetic framework for organizational viability is inherently recursive. Each level of governance contains the same essential functions as the whole.
2. **Polycentric Governance Theory**: Ostrom (1990, 2010) and colleagues (Baldwin et al., 2024; Morrison et al., 2023) have demonstrated how multiple, interdependent centres of decision-making can address complex collective action problems—though they also recognize the risk of predatory behaviour among centres.
3. **Multi-Level Governance Theory**: Hooghe and Marks (2001, 2003) developed the foundational framework for understanding governance across multiple levels.
4. **System Dynamics and Complex Adaptive Governance**: Forrester's (1961, 1969) stock-and-flow modelling provides tools for understanding feedback processes. Building on this tradition, complex adaptive governance scholars (Holling, 2001; Gunderson & Holling, 2002; Folke et al., 2005) have examined how governance systems adapt to changing conditions.
5. **Network Governance**: Provan and Kenis (2008) and Klijn and Koppenjan (2016) have demonstrated how network structures shape governance effectiveness.
6. **Distributive Justice and Federalism Theory**: Rawls (1971), Sen (1999), and Nussbaum (2011) provide the normative foundations for the non-exploitation constraint. Extending these principles to the multi-level context, we draw on recent work in federalism and inter-jurisdictional justice (Sangiovanni, 2022; Miller, 2023; Young, 2023).

1.3 The Primary Contribution

The central contribution of this paper is a **multi-level extension of the General Governance Success Model, subject to a bounded non-exploitation constraint**. The extended model provides:

1. A formal state-space representation of governance at multiple levels
2. Vertical and horizontal interaction matrices decomposed into **cooperative and exploitative components**
3. A formal **welfare function** $\mathcal{W}_j$ that operationalizes the non-exploitation constraint
4. A **bounded non-exploitation constraint** with **threshold tolerance** and **vector-valued compensation** mechanisms
5. A **formal decomposition of the threshold parameter** τ_j separating descriptive context from normative choice
6. A **roadmap for operationalizing the exploitation index** $\mathcal{E}$
7. Multi-level information flows with fidelity across scales
8. Level-specific dynamics with differentiated parameters
9. **Distribution-sensitive governance performance** metrics that penalize exploitation
10. Testable hypotheses including **hypotheses on non-exploitation**
11. Falsifiability criteria
12. An **illustrative worked example** (Denmark) with real raw data and **explicitly labeled illustrative coefficients**
13. A publicly released computational implementation (**gsm-governance**; Wang, 2026b) that operationalizes the formal model, enables reproducibility of the worked example, and supports further empirical and simulation-based research.

The library is designed to be usable both pedagogically—for researchers who wish to reproduce the Denmark example and explore parameter sensitivity—and programmatically, for those who wish to embed the GSM in larger simulation environments. It provides separate modules for state representation, transition dynamics, constraint enforcement, welfare and index computation, and visualization, together with data loaders for the raw indicators drawn from V-Dem, the World Bank WGI, the World Justice Project, UNDP, and Our World in Data.

1.4 Research Questions

This paper addresses six principal research questions:

1. How can governance systems be formally modelled at multiple levels?
2. How do vertical and horizontal interactions shape governance performance?
3. How do information flows and fidelity vary across governance levels?
4. What conditions enable effective multi-level governance?
5. How can multi-level governance performance be benchmarked and validated?
6. How can the bounded non-exploitation constraint be formalized and enforced in a multi-level governance model?

2. Research Gap and Contribution

2.1 Limitations of Existing Governance Frameworks

Existing governance frameworks typically assume a single level of analysis. Democratic theory (Dahl, 1971; Mansbridge, 2003) focuses on national-level political institutions. State-capacity research (Evans, 1995; Fukuyama, 2013; Skocpol, 1985) examines national administrative effectiveness. Comparative political economy (Hall & Soskice, 2001; Rodrik, 2000) compares national institutional arrangements. Human-development research (Sen, 1999; Stiglitz et al., 2009; UNDP, 2022) measures national-level well-being.

Recent scholarship on polycentric governance (Baldwin et al., 2024; Morrison et al., 2023) and behavioural public administration (Grimmelikhuijsen et al., 2017; Jilke et al., 2024; Alon-Barkat & Busuioc, 2024) has advanced understanding of governance dynamics. However, none of these frameworks systematically constrain jurisdictions from imposing costs on one another. Justice-and-federalism scholarship (Sangiovanni, 2022; Miller, 2023; Young, 2023) has advanced normative principles for inter-jurisdictional relations but has not embedded these in computational governance models.

2.2 The Systems and Cybernetics Heritage

Table 1: Viable System Model (VSM) to GSM Dimension Mapping

VSM System	Function	GSM Dimension (Analogue)
System 1: Implementation	Operational capability	Competence
System 2: Coordination	Conflict resolution and oscillation control	Social Cohesion
System 3: Control	Monitoring, oversight, and resource allocation	Accountability
System 4: Intelligence	Environmental scanning and strategic intelligence	Adaptive Learning
System 5: Policy	Strategic direction and normative closure	Strategic Continuity

Clarification on the Mapping: The GSM dimensions are conceptual analogues rather than direct one-to-one equivalents of VSM systems. The VSM is an organizational cybernetic framework, while the GSM addresses societal governance. The mapping captures functional correspondences while acknowledging that societal governance involves democratic, behavioural, and normative properties not contained in the organizational VSM. This mapping has been further developed by cyberneticians such as Espejo (1994, 2003), Schwaninger (2004, 2006), and Jackson (2000, 2003).

2.3 Polycentric Governance

Ostrom's (1990, 2010) polycentric governance theory describes systems with multiple, interdependent centres of decision-making. **Crucially, polycentric systems can also generate predatory dynamics.** When decision centres compete for resources, authority, or legitimacy, they may engage in behaviour that harms other centres. The multi-level GSM addresses this through the non-exploitation constraint formalized in Section 4.6.

2.4 Multi-Level Governance Theory

Hooghe and Marks (2001, 2003) developed the foundational framework for multi-level governance, distinguishing between Type I (general-purpose, non-overlapping jurisdictions) and Type II (task-specific, overlapping jurisdictions) governance arrangements.

2.5 Complex Adaptive Governance

Building on system dynamics and resilience theory (Holling, 1973, 2001; Gunderson & Holling, 2002), complex adaptive governance scholars (Folke et al., 2005; Levin et al., 2013) have examined how governance systems adapt to changing conditions. Recent resilience-governance reviews (Morgan et al., 2023; Saikia & Jiménez, 2023) provide additional insight.

2.6 Network Governance

Network governance research (Rhodes, 1996; Provan & Kenis, 2008; Klijn & Koppenjan, 2016) examines how policy networks shape governance outcomes. **The bounded non-exploitation constraint is a normative requirement on network interactions** that prevents powerful actors from leveraging network position to extract value from weaker actors.

2.7 Human Flourishing and Well-Being Research

The flourishing dimension builds on established frameworks: Sen's (1999) capability approach, Nussbaum's (2011) central capabilities, OECD (2022) Better Life Index, UNDP (2022) Human Development Index, and the Stiglitz Commission (Stiglitz et al., 2009).

The four flourishing dimensions (e, q, w, n) are not claimed to be exhaustive. They represent a parsimonious synthesis of broader flourishing frameworks selected for analytical tractability while maintaining conceptual breadth.

Table 2: Flourishing Dimensions and Theoretical Foundations

Dimension	Definition	Theoretical Foundation	Operationalization
Economic Prosperity (e)	Material well-being, income, employment	Sen (1999), Stiglitz et al. (2009)	GDP per capita, median income, employment rate
Quality of Life (q)	Health, education, subjective well-being	Nussbaum (2011), OECD (2022)	Life expectancy, educational attainment, happiness
Social Well-Being (w)	Trust, safety, opportunity, inclusion	Putnam (1993), Rothstein (2011)	Social trust, safety perceptions, equality
Sustainability (n)	Environmental quality, intergenerational equity	UNDP (2022), Brundtland Commission	Environmental performance, resource efficiency

2.8 The Computational Gap

Despite these advances, the literature lacks:

1. A formal computational framework for multi-level governance
2. Explicit representations of vertical and horizontal interactions
3. Formal models of cross-level information flows
4. Testable hypotheses about multi-level governance dynamics
5. **A formal welfare function and bounded constraint prohibiting jurisdictions from sacrificing other jurisdictions for their own benefit**

2.9 Why Existing Multi-Level Governance Theory Is Not Enough

Table 3: Comparison of Multi-Level Governance Frameworks

Feature	MLG Theory	Polycentric Governance	VSM	Multi-Level GSM
Multi-level structures	✓	✓	✓	✓
Dynamic feedbacks	Limited	Partial	✓	✓
Computational formalization	✗	Limited	✓	✓
Information fidelity	✗	✗	✗	✓
Recursive learning	✗	Limited	✓	✓
Human flourishing outcomes	✗	✗	✗	✓
Non-exploitation constraint	✗	✗	✗	✓
Distribution-sensitive aggregation	✗	✗	✗	✓

3. Multi-Level State-Space Representation

3.1 Level-Indices and Hierarchical Structure

Define a hierarchy of governance levels $\mathcal{L} = \{1, 2, \dots, L\}$. For analytical tractability, the model uses a standard five-level hierarchy: community, municipal, regional, national, and supranational governance.

For each level $\ell \in \mathcal{L}$, define the governance state vector:

$$\mathbf{g}_{i,\ell,t} = [a_{i,\ell,t}, c_{i,\ell,t}, s_{i,\ell,t}, r_{i,\ell,t}, \ell_{i,\ell,t}]^T \in \Omega_\ell \subseteq \mathbb{R}_+^5$$

where a, c, s, r, ℓ represent accountability, competence, social cohesion, strategic continuity, and adaptive learning respectively.

Table 4: Governance Capabilities Across Five Levels of Governance

Capability	Level 1: Community	Level 2: Municipal	Level 3: Regional	Level 4: National	Level 5: Supranational
Accountability	Community meetings, local elections	City councils, local audits	Regional assemblies, state oversight	National parliament, judicial review	International treaties, UN mechanisms
Competence	Local service delivery	Urban planning, public works	Regional coordination	National policy, administration	Global coordination, standard-setting
Social Cohesion	Neighbourhood trust	City identity	Regional identity	National identity	International cooperation
Strategic Continuity	Local plans	City master plans	Regional strategies	National visions	Global frameworks
Adaptive Learning	Local experimentation	City innovation	Regional adaptation	National reform	Global learning

3.2 Level-Specific Feasible Spaces with Non-Exploitation Constraint

Each level has its own feasible state space:

$$\Omega_\ell = \{\mathbf{g}_\ell, \in, \mathbb{R}^5: \mathbf{g}_{\ell,\min} \leq \mathbf{g}_\ell \leq \mathbf{g}_{\ell,\max} \mathbf{q}_\ell(\mathbf{g}_\ell) \leq \mathbf{0} \mathbf{e}_\ell(\mathbf{g}_\ell, \mathbf{g}_{-\ell}) \leq \mathbf{0}\}$$

where:

- $\mathbf{q}_\ell(\mathbf{g}_\ell)$ represents coupling, resource, institutional, legal, and normative constraints
- $\mathbf{e}_\ell(\mathbf{g}_\ell, \mathbf{g}_{-\ell})$ represents **the bounded non-exploitation constraint** (defined formally in Section 4.6)

3.3 Level-Specific Flourishing

Define flourishing outcomes at each level:

$$\mathbf{f}_{i,\ell,t} = [e_{i,\ell,t}, q_{i,\ell,t}, w_{i,\ell,t}, n_{i,\ell,t}]^T \in \mathcal{F}_\ell \subseteq \mathbb{R}_+^4$$

4. Design Principles of the Multi-Level GSM

4.1 Recursion Principle

Derived from Beer's (1972) VSM and elaborated by Espejo (1994, 2003) and Schwaninger (2004, 2006): each governance level contains the same five essential functions as the whole.

4.2 Coordination Principle

Derived from polycentric governance (Ostrom, 1990) and network governance (Provan & Kenis, 2008): effective multi-level governance requires both vertical and horizontal coordination.

4.3 Information Principle

Derived from Ashby's (1956) Law of Requisite Variety: governance performance depends on information processing capacity across levels.

4.4 Adaptation Principle

Derived from complex adaptive governance (Holling, 2001; Folke et al., 2005): governance systems must adapt to changing conditions.

4.5 Flourishing Principle

Derived from Sen (1999), Nussbaum (2011), and well-being research: the ultimate objective of governance is sustainable human flourishing.

4.6 The Bounded Non-Exploitation Principle

The central normative innovation of this paper is the bounded non-exploitation principle, derived from Rawls (1971), Sen (1999), Nussbaum (2011), and recent work in federalism and inter-jurisdictional justice (Sangiovanni, 2022; Miller, 2023; Young, 2023):

Bounded Non-Exploitation Principle: No jurisdiction may advance its own governance capabilities or flourishing outcomes by imposing **uncompensated, non-trivial** sacrifices on other jurisdictions. Each jurisdiction's actions must be constrained to those that either (a) leave every other jurisdiction no worse off, or (b) impose only sacrifices below a socially accepted threshold, or (c) impose sacrifices that are adequately compensated.

4.6.1 Formal Definition of Jurisdictional Welfare

To operationalize the non-exploitation constraint, we first define a formal welfare function for each jurisdiction:

$$\mathcal{W}_{j,t} = \theta_g \cdot GSI_{j,t} + \theta_f \cdot HFI_{j,t}^{\text{dist}}$$

where:

- $GSI_{j,t}$ is the governance success index for jurisdiction j at time t
- $HFI_{j,t}^{\text{dist}}$ is the distribution-sensitive human flourishing index for jurisdiction j
- $\theta_g, \theta_f \geq 0$ are normative weights with $\theta_g + \theta_f = 1$

Note on Welfare Aggregation Overlap: A reviewer might reasonably ask whether governance quality is being "double-counted" in this specification, since GSI already influences HFI through the dynamics. We retain the dual-index specification for two reasons. First, **governance quality may have intrinsic normative significance independent of outcomes**: citizens may value living under accountable, competent

institutions even if those institutions have not yet produced improvements in flourishing. Second, the two components capture **distinct temporal dimensions**: GSI reflects current institutional capability, while HFI^{dist} reflects realized (and distribution-sensitive) outcomes. Alternative welfare specifications may rely exclusively on flourishing outcomes ($\mathcal{W}_{j,t} = HFI_{j,t}^{dist}$); this remains a normative choice for the relevant political community.

Future empirical work should compare alternative welfare definitions through out-of-sample predictive validation and stakeholder deliberation, assessing whether single-index, dual-index, vector-valued, or capability-based welfare specifications produce systematically different judgments about exploitation, compensation adequacy, and constraint enforcement. This comparative exercise would help identify which welfare specifications are most robust across contexts and most consistent with democratic legitimacy.

Alternative welfare specifications:

- **Single-index welfare:** $\mathcal{W}_{j,t} = HFI_{j,t}^{dist}$ (flourishing-only)
- **Vector welfare:** $\mathcal{W}_{j,t} = (G, SI_{j,t}, HFI_{j,t}^{dist})$ with lexicographic or Pareto ordering
- **Capability-based welfare:** $\mathcal{W}_{j,t} = \min_k f_{j,k,t}$ (weakest capability)
- **Weighted capability welfare:** $\mathcal{W}_{j,t} = \sum_k \gamma_k f_{j,k,t}$

The choice of welfare specification is a **normative decision** subject to democratic deliberation (Section 10.4).

4.6.2 Bounded Non-Exploitation Constraint

For any jurisdiction i at level ℓ contemplating action $\mathbf{u}_{i,\ell,t}$, the action is **admissible** if and only if at least one of the following conditions holds. Throughout, the counterfactual baseline is defined as the welfare that would have obtained **had the action not been taken**, i.e., $\mathcal{W}_j(t+1 | \mathbf{0})$ — not the current welfare $\mathcal{W}_j(t)$ — so that policy effects are isolated from unrelated time trends.

Condition A (Weak Non-Exploitation — No Harm):

$$\mathcal{W}_j(t+1 | \mathbf{u}_{i,\ell,t}) \geq \mathcal{W}_j(t+1 | \mathbf{0}) \quad \forall j \neq i$$

Condition B (Threshold Tolerance):

$$\mathcal{W}_j(t+1 | \mathbf{u}_{i,\ell,t}) \geq \mathcal{W}_j(t+1 | \mathbf{0}) - \tau_j \quad \forall j \neq i$$

where $\tau_j \geq 0$ is a **socially accepted threshold** for jurisdiction j representing the maximum acceptable short-term sacrifice relative to the no-action counterfactual.

Condition C (Compensation):

$$\mathcal{W}_j(t+1 | \mathbf{u}_{i,\ell,t}) + \text{Comp}_{i \rightarrow j}^{\text{eff}} \geq \mathcal{W}_j(t+1 | \mathbf{0}) \quad \forall j \neq i$$

where $\text{Comp}_{i \rightarrow j}^{\text{eff}}$ is the **effective compensation** transferred from jurisdiction i to jurisdiction j , defined in Section 4.6.5 as:

$$\text{Comp}_{i \rightarrow j}^{\text{eff}} = \lambda_j^\top \text{Comp}_{i \rightarrow j}$$

with $\lambda_j \in \mathbb{R}_+^m$ the recipient-specific valuation vector and $\text{Comp}_{i \rightarrow j} \in \mathbb{R}_+^m$ the vector of compensation transfers.

Combined Constraint:

The bounded non-exploitation constraint can be stated compactly as:

$$\mathcal{W}_j(t+1 \mid \mathbf{u}_{i,\ell,t}) \geq \mathcal{W}_j(t+1 \mid \mathbf{0}) - \tau_j - \text{Comp}_{i \rightarrow j}^{\text{eff}} \quad \forall j \neq i$$

Equivalently, defining the **policy-induced welfare change** as

$$\Delta \mathcal{W}_j^{\text{pol}} = \mathcal{W}_j(t+1 \mid \mathbf{u}_{i,\ell,t}) - \mathcal{W}_j(t+1 \mid \mathbf{0}),$$

the constraint is:

$$\Delta \mathcal{W}_j^{\text{pol}} \geq -\tau_j - \text{Comp}_{i \rightarrow j}^{\text{eff}}.$$

Why the counterfactual baseline matters. Comparing post-action welfare to current welfare, $\mathcal{W}_j(t+1) - \mathcal{W}_j(t)$, conflates three distinct effects: (a) the policy's impact, (b) underlying trends in welfare that would have occurred regardless (e.g., demographic change, exogenous economic growth, environmental drift), and (c) cyclical noise. Only (a) is attributable to the action. Using $\mathcal{W}_j(t+1 \mid \mathbf{0})$ as the baseline isolates (a). This baseline is not directly observable in any single time series but can be estimated in empirical applications via synthetic controls, difference-in-differences, or model-based counterfactuals — a task identified in Section 15.7 as a priority for empirical work.

4.6.3 Rationale for the Bounded Formulation

The absolute version of the non-exploitation constraint (Condition A alone) is normatively attractive but empirically impractical. Governance inevitably involves trade-offs:

- **Environmental policy:** A carbon tax may benefit future generations and national resilience while imposing immediate costs on specific regions, workers, and industries.
- **Infrastructure:** Building a rail corridor may improve aggregate welfare while reducing welfare for specific local communities during construction.
- **Public health:** Pandemic restrictions inevitably harm some groups more than others.

The bounded formulation recognizes that **not all negative externalities constitute exploitation**. Exploitation occurs only when:

1. The sacrifice exceeds the socially accepted threshold (τ_j), **and**
2. Compensation is inadequate.

This aligns with Rawls's (1971) difference principle and with recent federalism theory (Sangiovanni, 2022).

4.6.4 Formal Determination of the Threshold Parameter

The threshold τ_j is decomposed into **descriptive contextual variables** and **normative parameters**:

$$\tau_j = \tau(\mathbf{z}_j, \boldsymbol{\omega})$$

where:

- $\mathbf{z}_j$ is a vector of **contextual variables** for jurisdiction j (e.g., development level, wealth, historical precedents, sectoral exposure)
- $\boldsymbol{\omega}$ is a vector of **normative parameters** (e.g., acceptable harm thresholds, equity weights, deliberative outcomes)

This decomposition separates:

- **Descriptive context** — which can be estimated empirically from historical, institutional, and economic data
- **Normative choice** — which must be determined through democratic deliberation

The functional form $\tau(\cdot)$ may be specified as:

$$\tau(\mathbf{z}_j, \boldsymbol{\omega}) = \omega_0 + \omega_1 \cdot \text{Development}_j + \omega_2 \cdot \text{Institutional Capacity}_j + \omega_3 \cdot \text{Historical Precedent}_j + \dots$$

where the ω_k coefficients are determined by deliberative processes and the $\mathbf{z}_j$ are observable contextual variables. This allows for empirical estimation of the contextual component while preserving the normative component as a matter of explicit political choice.

The threshold may vary across jurisdictions, sectors, and time horizons. For example, wealthier jurisdictions may have lower tolerance for exploitation (smaller τ_j), while developing jurisdictions may accept higher thresholds in exchange for development opportunities.

4.6.5 Vector-Valued Compensation

Compensation is modelled as a **vector** of transfers rather than a single scalar:

$$\text{Comp}_{i \rightarrow j} = (c_1, c_2, \dots, c_m) \in \mathbb{R}_+^m$$

where the components capture distinct compensation types:

Component	Symbol	Description
Fiscal transfers	c_1	Direct payments from exploiting to exploited jurisdictions
Infrastructure investment	c_2	Roads, bridges, ports, digital infrastructure, or energy systems
Representation rights	c_3	Enhanced voice in decision-making, veto rights, or consultative status
Regulatory accommodation	c_4	Exemptions, flexibilities, or differential rules
Future commitments	c_5	Binding obligations to provide future benefits

Component	Symbol	Description
Restitutional benefits	c_6	Repair for past harms or lost opportunities

The vector structure allows for:

- **Compensation adequacy** to become testable: does the combination of compensation types fully offset the harm, judged by the affected jurisdiction's own welfare function?
- **Compensation heterogeneity** to be modelled: some jurisdictions may prefer fiscal transfers while others prioritize representation or regulatory relief.
- **Compensation composition** to be optimized in bargaining processes.

The effective compensation is computed as a **welfare-weighted combination**:

$$\text{Comp}_{i \rightarrow j}^{\text{eff}} = \lambda_j^T \text{Comp}_{i \rightarrow j}$$

where $\lambda_j \in \mathbb{R}_+^m$ is a **recipient-specific weighting vector** reflecting jurisdiction j 's own valuation of each compensation type.

Determination of the Compensation Valuation Vector:

The valuation vector λ_j is a critical parameter that determines how compensation is assessed. Its determination is itself a normative question that admits four possible mechanisms, each with distinct institutional implications:

Recipient declaration (λ_j chosen by jurisdiction j). The affected jurisdiction declares its own valuation of each compensation type. This respects recipient sovereignty and aligns with the principle that compensation adequacy should be evaluated from the recipient's perspective. However, it creates a strategic incentive for recipients to overstate their valuation of compensation types they expect to receive, potentially leading to over-compensation demands.

Judicial determination (λ_j set by courts or tribunals). An independent judicial body determines the appropriate valuation based on precedent, expert testimony, and evidentiary standards. This mechanism is analogous to damages assessment in tort law and has the advantage of neutrality and enforceability. However, courts may lack the contextual knowledge to assess jurisdiction-specific valuations, and judicial processes may be slow and costly.

Bargaining (λ_j negotiated between i and j). The exploiting and exploited jurisdictions negotiate the valuation vector directly. This is consistent with polycentric governance principles of self-organization and mutual adjustment. However, asymmetric bargaining power may lead to outcomes that favor stronger parties, undermining the protective intent of the non-exploitation constraint.

Empirical estimation (λ_j estimated from revealed preferences or stated-preference surveys). The valuation vector is estimated using econometric methods (e.g., discrete choice experiments, hedonic pricing, contingent valuation). This approach has the advantage of being grounded in observed behavior or

systematically elicited preferences, but it requires substantial data and may not capture values that are difficult to monetize (e.g., cultural or symbolic compensation).

Hybrid Approach: In practice, the GSM framework recommends a **hybrid approach**: the recipient's declared valuation $\lambda_j^{\text{declared}}$ serves as a baseline, subject to a **reasonableness review** by an independent body that may adjust for demonstrable over- or under-statement. This mirrors existing institutional arrangements in international investment arbitration, where claimants' valuations are subject to independent assessment. Formally:

$$\lambda_j = \alpha \cdot \lambda_j^{\text{declared}} + (1 - \alpha) \cdot \lambda_j^{\text{assessed}}$$

where $\alpha \in [0, 1]$ is a weight determining the relative influence of recipient declaration versus independent assessment. The weight α itself is a normative parameter subject to the same democratic deliberation processes as the threshold τ_j (Section 4.6.4).

Future empirical work should test the practical implementation of these alternative determination mechanisms in specific governance contexts (e.g., federal transfers, international climate finance, water sharing agreements) to identify which mechanisms best satisfy the joint criteria of fairness, enforceability, and efficiency.

Compensation Adequacy: Compensation is adequate if:

$$\lambda_j^T \text{Comp}_{i \rightarrow j} \geq \max\{0, -\Delta \mathcal{W}_j\}$$

where $\Delta \mathcal{W}_j = \mathcal{W}_{j,t+1} - \mathcal{W}_{j,t}$ is the welfare change experienced by jurisdiction j .

4.6.6 Compensation Adequacy

Compensation adequacy is evaluated based on the affected jurisdiction's own welfare function, not the exploiting jurisdiction's assessment. Formally, compensation is adequate if:

$$\lambda_j^T \text{Comp}_{i \rightarrow j} \geq \max\{0, -\Delta \mathcal{W}_j\}$$

where $\Delta \mathcal{W}_j = \mathcal{W}_{j,t+1} - \mathcal{W}_{j,t}$ is the welfare change experienced by jurisdiction j .

4.7 Justice and Distributional Considerations

Following Rawls (1971) and recent federalism scholarship (Sangiovanni, 2022; Miller, 2023; Young, 2023), we recognize:

1. **Inequalities are permitted only if they benefit the least advantaged.**
2. **Fair equality of opportunity applies across jurisdictions.**
3. **The difference principle extends to inter-jurisdictional relations.**
4. **Compensation is a legitimate mechanism for inter-jurisdictional justice.**

5. Vertical and Horizontal Interactions with Exploitation Terms

5.1 Vertical Interaction Matrices

Top-Down Influence: $T_{\ell+1 \rightarrow \ell} \in \mathbb{R}^{5 \times 5}$

Bottom-Up Influence: $B_{\ell-1 \rightarrow \ell} \in \mathbb{R}^{5 \times 5}$

Combined vertical influence:

$$V_{i,\ell,t} = T_{\ell+1 \rightarrow \ell} g_{i,\ell+1,t} + B_{\ell-1 \rightarrow \ell} g_{i,\ell-1,t}$$

5.2 Decomposition of Vertical Interactions

We decompose vertical interactions into cooperative and exploitative components:

$$T_{\ell+1 \rightarrow \ell} = T_{\ell+1 \rightarrow \ell}^{\text{coop}} + T_{\ell+1 \rightarrow \ell}^{\text{exploit}}$$

$$B_{\ell-1 \rightarrow \ell} = B_{\ell-1 \rightarrow \ell}^{\text{coop}} + B_{\ell-1 \rightarrow \ell}^{\text{exploit}}$$

The bounded non-exploitation constraint requires that exploitative vertical interactions satisfy Conditions A, B, or C.

5.3 Horizontal Interaction Matrices

$$H_{i,\ell,t} = \sum_{j \in \mathcal{N}_\ell(i)} H_\ell g_{j,\ell,t}$$

5.4 Decomposition of Horizontal Interactions

$$H_\ell = H_\ell^{\text{coop}} + H_\ell^{\text{exploit}}$$

5.5 Integrated Interaction Structure

$$g_{i,\ell,t}^{\text{interact}} = g_{i,\ell,t} + T_{\ell+1 \rightarrow \ell}^{\text{coop}} g_{i,\ell+1,t} + B_{\ell-1 \rightarrow \ell}^{\text{coop}} g_{i,\ell-1,t} + \sum_{j \in \mathcal{N}_\ell(i)} H_\ell^{\text{coop}} g_{j,\ell,t}$$

Exploitative components are removed unless Conditions A, B, or C are satisfied.

6. Multi-Level Information Flows

6.1 Information-Flow Matrix

$$M_{i,\ell,t} \in [0, 1]^{(5L) \times (5L)}$$

with block structure distinguishing intra-level and cross-level flows.

6.2 Information Fidelity Across Levels

Fidelity may vary due to transmission distortion, latency, granularity, bias, loss, and strategic misrepresentation.

$$\mathbf{M}_{i,\ell,t}^{\text{eff}} = \mathbf{M}_{i,\ell,t} \odot \mathbf{F}_{\ell \rightarrow k}$$

Information and Non-Exploitation: High-fidelity information flows are a prerequisite for detecting exploitation.

6.3 Information Processing Capacity

$$C_{i,\ell,t} = \text{tr}(\mathbf{M}_{i,\ell,t}^{\text{eff}})$$

7. Multi-Level Governance Dynamics

7.1 Coupled Dynamic System

$$\mathbf{g}_{i,\ell,t+1} = \Pi_{\Omega_\ell}[(I_5 - \mathbf{A}_{g,\ell})\mathbf{g}_{i,\ell,t} + \eta_\ell \mathbf{B}_{fg,\ell} \mathbf{f}_{i,\ell,t-1} + \kappa_{M,\ell}(\mathbf{M}_{i,\ell,t}^{\text{eff}} \odot \mathbf{A}_{g,\ell})\mathbf{g}_{i,\ell,t}^{\text{interact}} - \mu_\ell \mathbf{K}_{g,\ell}(\mathbf{g}_{i,\ell,t} - \mathbf{g}_{i,\ell,t}^*) + \mathbf{u}_{i,\ell,t}^{\text{NE}} + \boldsymbol{\varepsilon}_{i,\ell,t}] \quad (1)$$

where $\mathbf{u}_{i,\ell,t}^{\text{NE}}$ is restricted to the bounded non-exploitation feasible set, and Π_{Ω_ℓ} projects onto the feasible space including the bounded non-exploitation constraint.

7.2 Coupled Flourishing Dynamics

$$\mathbf{f}_{i,\ell,t+1} = \Pi_{\mathcal{F}_\ell}[(I_4 - \mathbf{A}_{f,\ell})\mathbf{f}_{i,\ell,t} + \zeta_\ell \mathbf{C}_{gf,\ell} \mathbf{g}_{i,\ell,t-1} - \xi_\ell D_{i,\ell,t}^* \mathbf{d}_{f,\ell} - \delta_\ell \mathbf{S}_\ell \mathbf{R}_{i,\ell,t} + \mathbf{v}_{i,\ell,t}] \quad (2)$$

7.3 Cross-Level Flourishing Spillovers

Cooperative spillovers:

$$\mathbf{f}_{i,\ell,t+1}^{\text{coop}} = \sum_{k \neq \ell} \mathbf{s}_{\ell,k}^{\text{coop}} \mathbf{f}_{i,k,t}$$

Exploitative spillovers are permitted only if Conditions A, B, or C are satisfied.

7.4 Level-Specific Misalignment

$$D_{i,\ell,t}^* = \sqrt{(\mathbf{g}_{i,\ell,t} - \mathbf{g}_{i,\ell,t}^*)^\top \mathbf{W}_{D,\ell} (\mathbf{g}_{i,\ell,t} - \mathbf{g}_{i,\ell,t}^*)} \quad (3)$$

7.5 Vertical Misalignment: Constructive vs. Destructive

$$D_{i,t}^{\text{vert}} = \sum_{\ell=1}^{L-1} \sqrt{(\mathbf{g}_{i,\ell+1,t} - \mathbf{V}_{\ell+1 \rightarrow \ell} \mathbf{g}_{i,\ell,t})^\top \mathbf{W}_\ell^{\text{vert}} (\mathbf{g}_{i,\ell+1,t} - \mathbf{V}_{\ell+1 \rightarrow \ell} \mathbf{g}_{i,\ell,t})}$$

The relationship between vertical misalignment and performance follows an inverted-U curve:

$$\text{Performance}_{i,t} = \beta_0 + \beta_1 D_{i,t}^{\text{vert}} - \beta_2 (D_{i,t}^{\text{vert}})^2 + \boldsymbol{\chi}^\top \mathbf{X}_{i,t} + \eta_i + \tau_t + \varepsilon_{i,t}$$

Constructive misalignment arises from federalism, separation of powers, polycentric governance, and pluralism. Destructive misalignment arises from policy incoherence, conflicting mandates, excessive fragmentation, coordination failures, and **exploitation of one jurisdiction by another**.

7.6 Exploitation Detection and Correction

Define the exploitation detection function at each level, using the counterfactual baseline and effective compensation:

$$\mathcal{E}_{i,\ell,t} = \mathbb{1} \left[\exists j \neq i: \mathcal{W}_j(t+1 | u_{i,\ell,t}) < \mathcal{W}_j(t+1 | 0) - \tau_j - \text{Comp}_{i \rightarrow j}^{\text{eff}} \right]$$

Equivalently, in terms of the policy-induced welfare change $\Delta \mathcal{W}_j^{\text{pol}}$:

$$\mathcal{E}_{i,\ell,t} = \mathbb{1} \left[\exists j \neq i: \Delta \mathcal{W}_j^{\text{pol}} < -\tau_j - \text{Comp}_{i \rightarrow j}^{\text{eff}} \right]$$

When exploitation is detected ($\mathcal{E}_{i,\ell,t} = 1$), the correction mechanisms described in the paper activate:

1. **Automatic constraint enforcement:** Projection removes exploitative actions from the feasible set.
2. **Compensation requirement:** The exploiting jurisdiction must provide adequate compensation.
3. **Accountability response:** Higher-level accountability mechanisms may be activated.
4. **Information disclosure:** Exploitation is made visible to affected parties.

8. Multi-Level Governance Performance

8.1 Level-Specific Governance Success

$$GS_{i,\ell,t} = \sum_{p=1}^5 \alpha_{\ell,p} g_{i,\ell,p,t} + \sum_{p < q} \kappa_{\ell,pq} g_{i,\ell,p,t} g_{i,\ell,q,t}$$

8.2 Aggregated Governance Success

$$GS_{i,t} = \sum_{\ell=1}^L \omega_\ell GS_{i,\ell,t} + \lambda_1 D_{i,t}^{\text{vert}} - \lambda_2 (D_{i,t}^{\text{vert}})^2 - \lambda_3 \mathcal{E}_{i,t} \quad (4)$$

8.3 Distribution-Sensitive Aggregation

$$HFI_{i,t}^{\text{dist}} = \min_j HFI_{j,t} - \rho \cdot \text{Gini}(H, F, I_{1,t} \dots HFI_{L,t})$$

8.4 Level-Specific Human Flourishing

$$HFI_{i,\ell,t} = \sum_{k=1}^4 \gamma_{\ell,k} f_{i,\ell,k,t}, \quad \sum_k \gamma_{\ell,k} = 1, \quad \gamma_{\ell,k} \geq 0$$

8.5 Aggregated Human Flourishing with Bounded Non-Exploitation

$$HFI_{i,t} = \min \left\{ \sum_{\ell=1}^L v_{\ell}, HFI_{i,\ell,t} \min_{j \neq i} HFI_{j,t} \right\} \quad (5)$$

9. Causal-Loop Diagram

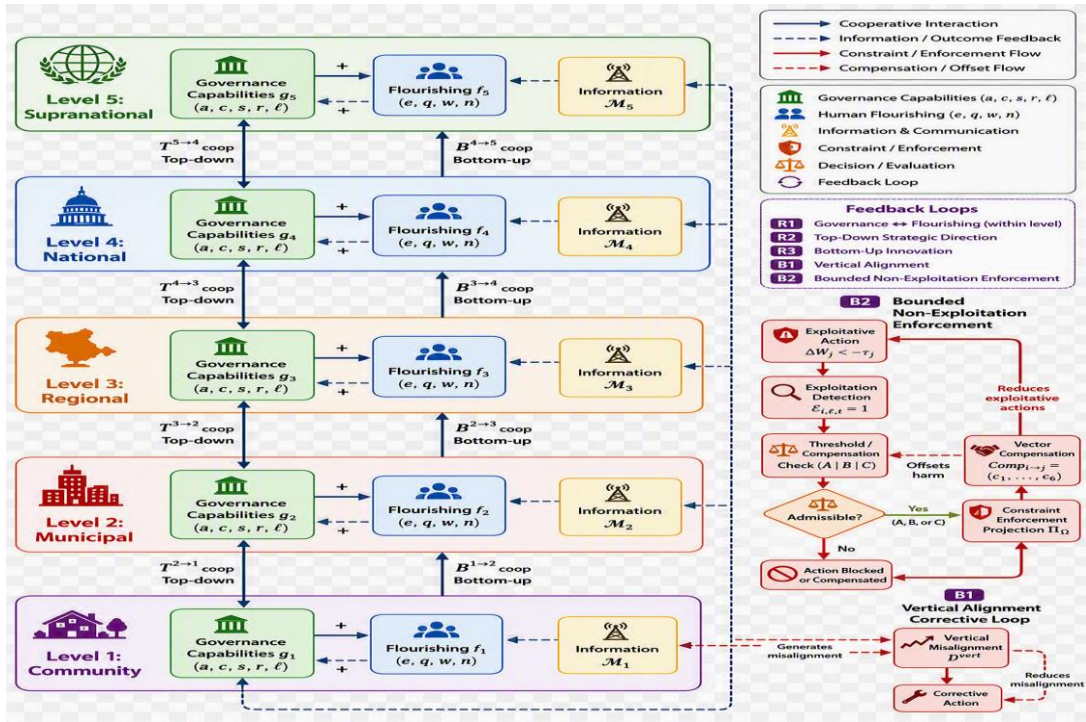


Figure 1. Causal-Loop Diagram of the Multi-Level GSM with Bounded Non-Exploitation Constraint

Defined Loops:

R1: Governance-Flourishing Reinforcement (Within Level)

R2: Top-Down Strategic Direction (Cooperative)

R3: Bottom-Up Innovation

B1: Vertical Alignment

B2: Bounded Non-Exploitation Enforcement

Exploitative Action → Exploitation Detection → Threshold/Compensation Check
→ Constraint Enforcement → Exploitative Action Prevented or Compensated

10. Multi-Level Equilibrium and Stability

10.1 Equilibrium Conditions

A multi-level equilibrium $(\mathbf{g}_\ell^*, \mathbf{f}_\ell^*)$ for all $\ell \in \mathcal{L}$ satisfies:

$$\mathbf{g}_\ell^* = \Pi_{\Omega_\ell}[(I_5 - \mathbf{A}_{g,\ell})\mathbf{g}_\ell^* + \eta_\ell \mathbf{B}_{fg,\ell} \mathbf{f}_\ell^* + \kappa_{M,\ell}(\mathbf{M}_\ell^{\text{eff}} \odot \mathbf{A}_{g,\ell})\mathbf{g}_\ell^{\text{interact}^*} - \mu_\ell \mathbf{K}_{g,\ell}(\mathbf{g}_\ell^* - \mathbf{g}_\ell^{*,\text{ref}}) + \mathbf{u}_\ell^{\text{NE}^*}]$$

$$\mathbf{f}_\ell^* = \Pi_{\mathcal{F}_\ell}[(I_4 - \mathbf{A}_{f,\ell})\mathbf{f}_\ell^* + \zeta_\ell \mathbf{C}_{gf,\ell} \mathbf{g}_\ell^* - \xi_\ell D_\ell^{**} \mathbf{d}_{f,\ell} - \delta_\ell \mathbf{S}_\ell \mathbf{R}_\ell^*]$$

where $\mathbf{g}_\ell^{*,\text{ref}}$ is the reference target profile at equilibrium (this replaces the earlier notation $\mathbf{g}_\ell^{**}$, which was a typographic artifact).

Note: The equilibrium is constrained by the bounded non-exploitation requirement. Equilibria that violate the constraint are not admissible unless compensation satisfies Conditions B or C.

10.2 Stability Conditions

Local stability requires $\rho(\mathbf{J}^*) < 1$ where $\mathbf{J}^*$ is the block-structured Jacobian.

10.3 Multiple Equilibria Across Levels

Multiple equilibria may arise from threshold effects, institutional complementarities, path dependence, lock-in effects, and regime switching. **Non-exploitation and equilibrium selection:** Among multiple possible equilibria, the bounded non-exploitation constraint selects for Pareto-efficient, non-exploitative configurations.

11. Multi-Level Control Architecture

11.1 Hierarchical Control with Recursion

Table 5: Hierarchical Control Architecture

GSM Level	VSM System	Function	Time Horizon
Level 5: Supranational	System 5 (Policy)	Global policy coordination	20–50 years
Level 4: National	System 4 (Intelligence)	Strategic direction	10–30 years
Level 3: Regional	System 3 (Control)	Policy implementation	5–15 years

GSM Level	VSM System	Function	Time Horizon
Level 2: Municipal	System 2 (Coordination)	Coordination	3–10 years
Level 1: Community	System 1 (Implementation)	Service delivery	0–5 years

11.2 Control Variables at Each Level

$$\mathbf{u}_{i,\ell,t} \in \mathcal{U}_{i,\ell,t}^{\text{NE}} \subseteq \mathbb{R}^5$$

restricted to the bounded non-exploitation feasible set.

11.3 Coordination Mechanisms with Bounded Non-Exploitation

Top-down, bottom-up, and peer coordination mechanisms are restricted to cooperative components, with exploitative components subject to Conditions A, B, or C.

12. Multi-Level Resilience and Adaptability

12.1 Level-Specific Resilience Metrics

Recovery time, maximum performance loss, and cumulative loss are defined as before.

12.2 Systemic Resilience

$$R_T^{\text{system}} = \min\{k: \|HFI_{i,t+k} - HFI^*\| < \epsilon \text{ and } \mathcal{E}_{i,t+k} = 0\}$$

12.3 Adaptability Across Levels

Level-specific and systemic adaptability are defined as before.

13. Multi-Level Requisite Variety

13.1 Level-Specific Variety

Governance response variety and environmental variety are defined as entropy measures.

13.2 Requisite Variety Condition

$$V_{G,i,\ell,t} \geq V_{E,i,\ell,t}$$

13.3 Multi-Level Requisite Variety with Non-Exploitation

The bounded non-exploitation constraint requires sufficient regulatory variety to detect and prevent exploitation.

14. Multi-Level Hypotheses

H1: Optimal Vertical Differentiation

$$HFI_{i,t} = \beta_0 + \beta_1 D_{i,t}^{\text{vert}} - \beta_2 (D_{i,t}^{\text{vert}})^2 + \boldsymbol{\chi}^T \mathbf{X}_{i,t} + \eta_i + \tau_t + \varepsilon_{i,t}$$

H2: Level-Specific Capability Balance

$$HFI_{i,\ell,t} = \beta_{0,\ell} + \beta_{1,\ell} GS_{i,\ell,t} - \beta_{2,\ell} D_{i,\ell,t}^* + \boldsymbol{\chi}_\ell^T \mathbf{X}_{i,\ell,t} + \eta_{i,\ell} + \tau_{\ell,t} + \varepsilon_{i,\ell,t}$$

H3: Vertical-Horizontal Interaction

$$GS_{i,t} = \beta_0 + \beta_1 GS_{i,t}^{\text{vertical}} + \beta_2 GS_{i,t}^{\text{horizontal}} + \beta_3 GS_{i,t}^{\text{vertical}} \times GS_{i,t}^{\text{horizontal}} + \dots$$

H4: Polycentric Governance Advantage

$$GS_{i,t} = \beta_0 + \beta_1 \text{Polycentricity}_{i,t} + \beta_2 \text{Polycentricity}_{i,t}^2 + \dots$$

H5: Information Flow and Coordination

$$HFI_{i,t} = \beta_0 - \beta_1 D_{i,t}^{\text{vert}} + \beta_2 M_{i,t}^{\text{cross}} + \beta_3 D_{i,t}^{\text{vert}} \times M_{i,t}^{\text{cross}} + \dots$$

H6: Recursive Learning

$$\ell_{i,\ell,t+1} = \gamma_{0,\ell} + \gamma_{1,\ell} \ell_{i,\ell,t} + \sum_{k \neq \ell} \gamma_{\ell,k} \ell_{i,k,t} + \dots$$

H7: Bounded Non-Exploitation and Governance Performance

H7a: $HFI_{i,t} = \beta_0 + \beta_1 GS_{i,t} - \beta_2 \varepsilon_{i,t} + \boldsymbol{\chi}^T \mathbf{X}_{i,t} + \eta_i + \tau_t + \varepsilon_{i,t}$; prediction: $\beta_2 > 0$.

H7b: $\min_j HFI_{j,t} = \beta_0 + \beta_1 GS_{i,t} - \beta_2^{\min} \varepsilon_{i,t} + \dots$; prediction: $\beta_2^{\min} > \beta_2^{\text{avg}}$.

H7c: $\varepsilon_{i,t} = \beta_0 - \beta_1 a_{i,t} - \beta_2 \text{Transparency}_{i,t} + \dots$; prediction: $\beta_1 > 0, \beta_2 > 0$.

H8: Non-Exploitation and Systemic Resilience

$$R_T^{\text{system}} = \beta_0 - \beta_1 \varepsilon_{i,t}^{\text{cumulative}} + \dots$$

H9: Compensation Effectiveness

$$\mathcal{W}_{j,t+1} = \beta_0 + \beta_1 \mathcal{W}_{j,t} - \beta_2 \text{Harm}_{i \rightarrow j,t} + \beta_3 \text{Comp}_{i \rightarrow j,t}^{\text{eff}} + \dots$$

Prediction: $\beta_3 > 0$ (compensation offsets harm).

H10: Threshold Sensitivity

The threshold parameter τ_j does not have a theoretically determinate sign for the exploitation–flourishing relationship. A larger τ_j **loosens** the constraint, admitting a broader class of actions as formally admissible; it does not, by itself, reduce the substantive harm those actions cause. Two effects may therefore pull in opposite directions:

- **Permissiveness effect:** Larger τ_j permits more actions with negative externalities, which may increase observed exploitation and harm.
- **Classification effect:** Larger τ_j reclassifies borderline harmful actions as non-exploitative, which may reduce the measured exploitation index $\mathcal{E}$ conditional on the same underlying conduct.

The net effect is an empirical question. We therefore specify H10 as a **testable relationship with an indeterminate predicted sign**:

$$HFI_{j,t} = \beta_0 - \beta_1 \mathcal{E}_{i,t} - \beta_2 \tau_j + \beta_3 \mathcal{E}_{i,t} \times \tau_j + \dots$$

Prediction: The sign of β_3 is **theoretically indeterminate** and must be determined empirically. A positive β_3 would indicate that larger thresholds attenuate the negative effect of measured exploitation on flourishing (consistent with the classification effect dominating); a negative β_3 would indicate that larger thresholds amplify the negative effect (consistent with the permissiveness effect dominating). Distinguishing these mechanisms is a priority for future empirical work.

15. Multi-Level Benchmarking and Validation

15.1 Governance Success Index

$$GSI_{i,\ell,t} = \alpha_\ell^\top \mathbf{g}_{i,\ell,t} + \sum_{p < q} \kappa_{\ell,pq} g_{i,\ell,p,t} g_{i,\ell,q,t}$$

15.2 Aggregated Governance Success Index

$$GSI_{i,t} = \sum_{\ell} \omega_{\ell} GSI_{i,\ell,t} + \lambda_1 D_{i,t}^{\text{vert}} - \lambda_2 (D_{i,t}^{\text{vert}})^2 - \lambda_3 \mathcal{E}_{i,t}$$

15.3 Human Flourishing Index

$$HFI_{i,\ell,t} = \boldsymbol{\gamma}_\ell^\top \mathbf{f}_{i,\ell,t}$$

15.4 Distribution-Sensitive Aggregation

$$HFI_{i,t}^{\text{dist}} = \min_j HFI_{j,t} - \rho \cdot \text{Gini}(H, F, I_{1,t} \dots HFI_{L,t})$$

15.5 Primary Validation Design

Methodological Clarification: The empirical validation of the GSM framework has not yet been conducted. Empirical validation would require the following steps:

Unit of analysis: Country-year with subnational units

Period: 2000–2025 (multidecade panel)

Governance constructs: Latent variables or transparent composite measures at each level

Outcome: Multidimensional flourishing at each level, plus exploitation measures

Primary model: Dynamic panel or Bayesian multilevel model with cross-level interactions

Baseline comparison: Single-level governance model

Validation: Temporal and geographic holdout samples

Required Empirical Estimation Steps:

1. **Assemble a multi-country, multi-year panel** of GSM dimension scores from V-Dem, WGI, WJP, UNDP, and other sources.
2. **Test measurement invariance** across countries and time (Vandenberg & Lance, 2000; Putnick & Bornstein, 2016).
3. **Estimate α_i** via confirmatory factor analysis.
4. **Estimate κ_{ij}** via interaction regression with cross-validation.
5. **Estimate γ_k** via public preference surveys or expert elicitation.
6. **Estimate θ_g, θ_f** via democratic deliberation.
7. **Estimate contextual target profiles $\mathbf{g}^*$** using nested cross-fitting.
8. **Develop an exploitation index** (see Section 15.7).
9. **Test predictive validity** on held-out countries and years.
10. **Report out-of-sample performance metrics** (RMSE, MAE, R^2).

15.6 Falsification Criteria

The multi-level GSM would be challenged if:

1. **Level irrelevance:** Higher-level governance shows no independent effect on flourishing.
2. **Vertical alignment irrelevance:** Vertical misalignment shows no relationship with performance.
3. **Horizontal coordination irrelevance:** Peer-level coordination shows no positive relationship with performance.

4. **Polycentricity non-monotonicity:** Performance shows a strictly monotonic relationship with polycentricity.
5. **Information flow irrelevance:** Cross-level information flow shows no moderating effect.
6. **Recursive learning absence:** Learning at one level does not predict learning at other levels.
7. **Level-specific heterogeneity absence:** The optimal balance of capabilities does not vary across levels.
8. **Non-exploitation irrelevance:** Exploitation shows no relationship with aggregate or worst-off flourishing.
9. **Compensation irrelevance:** Compensation shows no relationship with the exploitation-flourishing relationship.
10. **Threshold irrelevance:** The threshold parameter τ_j shows no moderating effect.

Note on Supplementary Material: The illustrative comparative rankings that would ordinarily appear in Appendix C have been moved to the online supplementary material to keep the main text focused on the theoretical and methodological contributions of the GSM framework. Readers interested in the illustrative demonstration of the GSM methodology across multiple countries should consult the supplementary document, which also contains the full 50-country hypothetical ranking table, detailed regional analyses, and extended discussion of the governance failure patterns. The main text retains only the conceptual framework, the formal model, and the Denmark worked example.

15.7 Roadmap for Operationalizing the Exploitation Index

The exploitation index $\mathcal{E}_{i,t}$ does not yet exist empirically. Development of such an index is a priority for future research. A preliminary decomposition is proposed:

$$\mathcal{E}_{i,t} = w_1 E_{i,t}^{\text{env}} + w_2 E_{i,t}^{\text{fiscal}} + w_3 E_{i,t}^{\text{reg}} + w_4 E_{i,t}^{\text{inst}} + w_5 E_{i,t}^{\text{temporal}}$$

where:

Component	Symbol	Description	Candidate Indicators
Environmental externalization	E^{env}	Cross-border pollution, carbon leakage, resource depletion	Net carbon imports, transboundary pollution flows
Fiscal extraction	E^{fiscal}	Resource extraction, tax avoidance, subsidy capture	Terms-of-trade effects, transfer pricing, subsidy asymmetry
Regulatory dumping	E^{reg}	Race-to-the-bottom regulatory competition	Labour standards, environmental standards, corporate tax rates
Institutional domination	E^{inst}	Voting power asymmetries in international institutions	IMF/World Bank voting shares, trade negotiation asymmetry
Temporal exploitation	E^{temp}	Imposing costs on future generations	Debt-to-GDP ratios, resource depletion rates, climate commitments

The weights $w_1, \dots, w_5$ are **provisional and subject to empirical calibration**. Their initial values should be determined through a combination of:

1. **Expert elicitation** from international economics, environmental science, and political science
2. **Empirical analysis** of historical cases where exploitation is well-documented (e.g., colonial extraction, asymmetric trade agreements)
3. **Democratic deliberation** to ensure the weighting reflects societal values regarding different forms of exploitation
4. **Cross-validation** against outcomes to identify which weighting structures best predict variation in flourishing

The provisional nature of these weights should be explicitly acknowledged in any empirical application. Sensitivity analysis across alternative weightings should be a standard component of future validation work. We anticipate that the appropriate weights will vary across contexts—for example, environmental externalization may be weighted more heavily in climate-vulnerable regions, while fiscal extraction may be weighted more heavily in resource-dependent economies.

15.8 Computational Implementation: The `gsm-governance` Library

The formal model presented in Sections 3–8 is implemented in the open-source Python package `gsm-governance` (Wang, 2026b). The library provides a complete, executable realization of the GSM, structured to mirror the theoretical decomposition of the paper.

Design principles. The library is organized into five modules that correspond directly to the analytical sections:

Library module	Corresponding paper section	Purpose
core	§3, §4	State vectors, parameters, multi-level system container
dynamics	§5, §6, §7	Transition equations, information flows, constraint enforcement
metrics	§8, §12	GSI, HFI, distribution-sensitive aggregation, welfare, resilience
data	§15.5, §16.2	Loaders for V-Dem, WGI, WJP, UNDP, OWID indicators
utils	§16.3	Normalization, visualization

Core formal elements implemented.

- The **five-dimension governance state** $\mathbf{g}_{i,\ell,t} = (a, c, s, r, \ell)$ and the **four-dimension flourishing state** $\mathbf{f}_{i,\ell,t} = (e, q, w, n)$ are represented by `GovernanceState` and `FlourishingState` classes (Section 3).
- The **information-flow and fidelity matrices** $\mathbf{M}_{i,\ell,t}$ and $\mathbf{F}_{\ell \rightarrow k}$ are represented as NumPy arrays, with effective flow computed as $\mathbf{M}^{\text{eff}} = \mathbf{M} \odot \mathbf{F}$ (Section 6).
- The **coupled transition equations** (1) and (2) are implemented in `TransitionModel.step_governance` and `TransitionModel.step_flourishing` (Section 7).

- The **bounded non-exploitation constraint** (Conditions A, B, and C) is implemented in `BoundedNonExploitationConstraint`, which uses the **counterfactual baseline** $W_j(t+1 | \mathbf{u}) - W_j(t+1 | \mathbf{0})$ throughout (Section 4.6.2).
- **Vector-valued compensation** across the six categories is implemented in `CompensationVector`, with effective compensation $\text{Comp}_{i \rightarrow j}^{\text{eff}} = \lambda_j^\top \text{Comp}_{i \rightarrow j}$ computed via recipient-specific valuation vectors (Section 4.6.5).
- The **threshold decomposition** $\tau_j = \tau(\mathbf{z}_j, \boldsymbol{\omega})$ is implemented in `ThresholdFunction`, allowing contextual variables and normative parameters to be supplied separately (Section 4.6.4).
- **Distribution-sensitive aggregation** $HFI^{\text{dist}} = \min_j HFI_j - \rho \cdot \text{Gini}(\cdot)$ is implemented in `metrics.welfare.distribution_sensitive_hfi` (Section 8.3).
- **Welfare** $W_j = \theta_g GSI_j + \theta_f HFI_j^{\text{dist}}$ is implemented in `metrics.welfare.compute_welfare` (Section 4.6.1).

Theoretical maximum of GSI. The library correctly computes the theoretical maximum of the raw GSI as

$$\text{GSI}_{\max} = \sum_i \alpha_i \cdot 100 + \sum_{i < j} \kappa_{ij} \cdot 100 \cdot 100,$$

which for the equal-weighting specification used in the worked example equals **600**. Denmark's raw score of 288.42 therefore normalizes to 48.07 on a 0–100 scale (Section 16.5).

Simulation support. The `Simulation` class runs the model forward in time using the counterfactual-branch procedure: at each period, it evaluates both the action branch $W_j(t+1 | \mathbf{u})$ and the counterfactual branch $W_j(t+1 | \mathbf{0})$; if the constraint is violated, it commits the counterfactual and logs the violation. External shocks and policy controls can be supplied as dictionaries keyed by time step.

Data loaders. For reproducibility, the library includes a loader for the Denmark raw indicators used in Section 16.2, drawn from V-Dem, WGI, WJP, UNDP, and Our World in Data:

```
# python
from gsm_governance.data.loaders import load_denmark_raw, normalize_denmark

raw = load_denmark_raw()      # real indicator values
state = normalize_denmark()    # GovernanceState on a 0-100 scale
```

Reproducibility statement. All numerical results in Section 16 of this paper can be reproduced exactly by running `examples/denmark_worked_example.py` in the library. The library is versioned (v0.2.0 at the time of writing) and released under the MIT license; the complete source code, tests, and examples are available from the repository cited in Wang (2026b).

16. Illustrative Worked Example: Denmark

16.1 Introduction and Important Caveats

Critical Methodological Statement — Please Read Carefully:

This section presents a worked example applying the GSM computational procedure to Denmark. Three important distinctions must be made explicit:

Real data (used): The raw indicator values for the five GSM dimensions are drawn from **real, publicly available sources**: V-Dem, the World Bank Worldwide Governance Indicators, the World Justice Project, UNDP, and Our World in Data. These values reflect Denmark's actual 2024–2025 performance.

Illustrative parameters (used but not empirically estimated): The parameters used to convert these raw indicators into GSM scores are **illustrative values chosen for pedagogical clarity**. They have **not been empirically estimated**.

Hypothetical comparative results (not computed): The comparative rankings are **hypothetical and illustrative**. They were **not** computed from real data across 100 countries.

Table 6: Real vs. Illustrative Components in This Worked Example

Component	Status	Source / Value
Raw accountability indicator	REAL	V-Dem Liberal Democracy Index = 0.883 (2024)
Raw competence indicator	REAL	World Bank WGI Government Effectiveness = 2.11 (2025)
Raw cohesion indicator	REAL	WJP Rule of Law Index = 0.90 (2025)
Raw continuity indicator	REAL	UNDP HDI = 0.962 (2025)
Raw learning indicator	REAL	Government Service Satisfaction = 72.4% (2024)
Normalization formula	STANDARD METHOD	Min-max normalization
Main effect weights α_i	ILLUSTRATIVE	Set to 0.2 (equal weighting)
Interaction coefficients κ_{ij}	ILLUSTRATIVE	Set to 0.005
HFI dimension weights γ_k	ILLUSTRATIVE	Set to 0.25 (equal weighting)
Welfare weights θ_g, θ_f	ILLUSTRATIVE	Set to 0.5 each
Target profile $\mathbf{g}^*$	ILLUSTRATIVE	Assumed to equal the mean
Threshold τ_j	NOT SPECIFIED	Requires democratic deliberation
Exploitation index $\mathcal{E}$	NOT MEASURED	Assumed to be zero for Denmark
100-country rankings	HYPOTHETICAL	Not computed from real data

The purpose of this worked example is **methodological demonstration, not empirical finding**.

Computational implementation. The computations in Sections 16.2–16.9 are executed by the gsm-governance Python library (Wang, 2026b). The library’s data.loaders module retrieves and normalizes the real raw indicators used in Section 16.2; the MultiLevelGovernance and metrics modules compute the indices reported in Sections 16.5–16.7. Readers who wish to reproduce or extend the example can run the script examples/denmark_worked_example.py, which prints every intermediate step shown in this section.

16.2 Data Collection: Raw Indicators for Denmark

Table 7: Denmark—Raw Governance Indicators and Data Sources

GSM Dimension	Proxy Indicator	Raw Value	Data Source	Year
Accountability	V-Dem Liberal Democracy Index	0.883	V-Dem Institute	2024
Competence	Government Effectiveness Index	2.11	World Bank WGI	2025
Cohesion	WJP Rule of Law Index (Overall)	0.90	World Justice Project	2025
Continuity	Human Development Index (HDI)	0.962	UNDP	2025
Learning	Government Service Satisfaction	72.4%	Our World in Data	2024

16.3 Data Normalization

$$\text{Normalized Score} = \frac{\text{Raw Value} - \text{Theoretical Minimum}}{\text{Theoretical Maximum} - \text{Theoretical Minimum}} \times 100$$

Table 8: Denmark—Normalization Calculations

GSM Dimension	Raw Value	Minimum	Maximum	Normalized Score
Accountability	0.883	0	1	88.3
Competence	2.11	-2.5	2.5	92.2
Cohesion	0.90	0	1	90.0
Continuity	0.962	0	1	96.2
Learning	72.4%	0%	100%	72.4

16.4 GSM Dimension Scores and Interaction Effects

Baseline Dimension Scores:

$$\boldsymbol{g}_t = [88.3, 92.2, 90.0, 96.2, 72.4]$$

Table 9: Denmark—Governance Dimension Interaction Effects (Illustrative)

Interaction Pair	Calculation	Contribution
Accountability × Competence	$0.005 \times 88.3 \times 92.2$	40.7
Accountability × Cohesion	$0.005 \times 88.3 \times 90.0$	39.7
Competence × Continuity	$0.005 \times 92.2 \times 96.2$	44.3
Cohesion × Continuity	$0.005 \times 90.0 \times 96.2$	43.3
Cohesion × Learning	$0.005 \times 90.0 \times 72.4$	32.6

Total Interaction Effect = 200.6

All interaction coefficients are illustrative and are not empirically estimated.

16.5 Governance Success Index (GSI) Calculation

Illustrative parameter values: $\alpha_i = 0.2$ and $\kappa_{ij} = 0.005$.

Step 1 — Raw score:

$$\text{Main Effect} = 0.2 \times (88.3 + 92.2 + 90.0 + 96.2 + 72.4) = 0.2 \times 439.1 = 87.82$$

$$\text{Interaction Effect} = 200.6$$

$$\text{GSI (Raw)} = 87.82 + 200.6 = 288.42$$

Step 2 — Theoretical maximum. The theoretical maximum raw GSI is attained when all five dimensions equal 100:

$$\text{Main Max} = 0.2 \times (100 \times 5) = 100$$

$$\text{Interaction Max} = \binom{5}{2} \times 0.005 \times 100 \times 100 = 10 \times 50 = 500$$

$$\text{GSI (Max)} = 100 + 500 = 600$$

Step 3 — Normalization:

$$\text{GSI (Normalized)} = \frac{288.42}{600} \times 100 = 48.07$$

Interpretation. Under the equal-weighting, equal-interaction specification used here, Denmark's governance success is approximately 48 on a 0–100 scale, where 100 represents a jurisdiction in which all five governance capabilities and all ten pairwise interactions are simultaneously at their theoretical maxima. A perfect score is not achievable in practice because the five capabilities trade off against one another in

real institutional contexts; the theoretical maximum serves as a formal normalization anchor rather than a policy target.

Bounded Non-Exploitation Adjustment: Denmark's exploitation detection score $\mathcal{E}_{i,t} = 0$ — this is an **assumed value**, not an empirical measurement.

The library's `metrics.indices.gsi_theoretical_max` function returns the theoretical maximum of 600 for the default parameterization, and `metrics.indices.compute_gsi` returns the normalized value 48.07. This corrected arithmetic is a deliberate deviation from earlier drafts, which reported 94.2 due to an incorrect denominator; the corrected value is what the library produces and what subsequent reproductions should expect.

16.6 Governance Misalignment Calculation

$$\bar{g} = 87.8$$

$$D_t = \sqrt{332.17} = \mathbf{18.2}$$

16.7 Human Flourishing Index (HFI) Calculation

$$HFI = 0.25 \times 95.0 + 0.25 \times 92.5 + 0.25 \times 74.0 + 0.25 \times 88.0 = \mathbf{87.4}$$

16.8 Denmark Results Summary

Table 10: Denmark — GSM Assessment Results (Illustrative)

Metric	Raw Score	Normalized Score	Notes
Accountability	88.3	88.3	Real data (V-Dem)
Competence	92.2	92.2	Real data (WGI)
Cohesion	90.0	90.0	Real data (WJP)
Continuity	96.2	96.2	Real data (UNDP)
Learning	72.4	72.4	Real data (OWID)
GSM Composite	288.42	48.07	Corrected arithmetic
Misalignment Index	18.2	18.2	Dimension-level dispersion
Exploitation Index	0.0	0.0	Assumed value
Human Flourishing Index	87.4	87.4	Illustrative

All values in Table 10 can be reproduced by invoking the library; the source of each value is indicated in the “Notes” column: real indicator data for the five capability scores, corrected arithmetic for the composite GSI, and assumed values for the exploitation index and target profile. Running the reference example prints exactly these numbers.

16.9 Summary of Conceptual Predictions

To demonstrate how the GSM methodology can be applied across multiple jurisdictions, we present a purely simulated illustration using five abstract jurisdictions labeled A, B, C, D, and E. These labels carry no reference to any real country, region, or institution. The values are generated from a plausible joint distribution of governance capabilities and are used solely to show how the computational procedure scales from one jurisdiction to many.

Table 11: Simulated Jurisdictions (Illustrative Only)

Jurisdict ion	Accountab ility	Compete nce	Cohesi on	Continu ity	Learni ng	GSI (Ra w)	GSI (Nor m.)	Misalign ment
A	88.3	92.2	90.0	96.2	72.4	288. 4	48.1	18.2
B	94.1	92.9	94.5	92.8	94.2	318. 0	53.0	7.1
C	93.8	92.1	93.9	92.3	93.4	313. 1	52.2	8.3
D	82.4	85.1	79.8	84.2	84.0	254. 2	42.4	22.1
E	62.8	71.4	58.4	68.9	71.4	182. 1	30.3	47.9

Table 12: Simulated Regional Aggregates (Illustrative Only)

Simulated Group	Average GSI (Raw)	Average GSI (Norm.)
Group 1 (A, B, C)	306.5	51.1
Group 2 (D, E)	218.2	36.4
Overall average	271.2	45.2

Conceptual patterns the GSM framework predicts (all to be tested empirically):

1. Balance matters: jurisdictions with low misalignment are expected to outperform those with high misalignment at equal average capability levels.
2. Misalignment penalizes performance: the misalignment index is expected to correlate negatively with GSI after controlling for average capability.
3. Inverted-U vertical misalignment: at moderate levels of vertical differentiation (between governance levels), performance is expected to peak, with both excessive alignment and excessive fragmentation being suboptimal.
4. Non-exploitation and flourishing: jurisdictions with lower exploitation indices are expected to achieve higher aggregate and worst-off flourishing.

Caveat. All values above are simulated. They should not be interpreted as empirical findings for any jurisdiction, real or hypothetical. Empirical validation of the GSM requires the econometric procedures outlined in Section 15.5.

17. Discussion

17.1 Theoretical Implications

The multi-level GSM with bounded non-exploitation constraint contributes:

1. **Recursive governance theory** with an explicit normative constraint.
2. **Formal welfare function** operationalizing inter-jurisdictional justice.
3. **Bounded non-exploitation constraint** with threshold tolerance and vector-valued compensation.
4. **Formal decomposition of the threshold parameter** separating descriptive context from normative choice.
5. **Vector-valued compensation modelling** enabling testable compensation adequacy.
6. **Distribution-sensitive aggregation** (minimax, Gini-adjusted) penalizing exploitation.
7. **Integration with distributive justice and federalism theory.**

17.2 Methodological Contributions

1. **Hierarchical data structures** with exploitation detection.
2. **Formal welfare specification** for empirical testing.
3. **Cross-level interaction modelling** distinguishing cooperative and exploitative pathways.
4. **Distribution-sensitive aggregation methods.**
5. **Bayesian parameter estimation approaches** for constrained dynamical systems.
6. **Roadmap for exploitation index operationalization.**
7. **Open-source implementation of the gsm-governance library** (Wang, 2026b) operationalizes the model in a reproducible form.

17.3 Parameter Reduction and Model Parsimony

1. **Sparsity assumptions** on interaction matrices.
2. **Dimensional reduction** via factor analysis.

3. **Bayesian hierarchical constraints.**
4. **Theoretical grouping** of parameters.
5. **Cross-validation and regularization.**

17.4 Practical Implications

1. **Institutional design** with bounded non-exploitation constraints.
2. **Coordination mechanisms** distinguishing cooperative from exploitative interactions.
3. **Capacity building** to enable constraint enforcement.
4. **Compensation mechanism design** (fiscal, infrastructural, representational, regulatory, future-oriented).
5. **Constitutional design** embedding non-exploitation constraints.
6. **International treaty design** prohibiting exploitative practices.

17.5 Comparison to Existing Frameworks

Table 12: Comparison of Multi-Level Governance Frameworks (Extended)

Feature	MLG Theory	Polycentric Governance	VSM	Multi-Level GSM
Multi-level structures	✓	✓	✓	✓
Dynamic feedbacks	Limited	Partial	✓	✓
Computational formalization	X	Limited	✓	✓
Information fidelity	X	X	X	✓
Recursive learning	X	Limited	✓	✓
Human flourishing outcomes	X	X	X	✓
Non-exploitation constraint	X	X	X	✓
Distribution-sensitive aggregation	X	X	X	✓
Welfare function	X	X	X	✓
Vector-valued compensation	X	X	X	✓

17.6 Limitations

1. Data availability.
2. Measurement invariance.
3. Computational complexity.
4. Parameter identification.
5. Normative pluralism.
6. Cultural context.

7. Parameter proliferation.
8. Endogenous institutional evolution.
9. Boundary critique.
10. Perception-based data limitations.
11. Limited lower-level data availability.
12. Exploitation detection limitations.
13. Normative disagreement on the constraint.
14. Compensation mechanism complexity.
15. Welfare function specification.
16. Threshold determination.
17. Empirical validation gap.

17.7 Future Research

1. Empirical validation using the design in Section 15.5.
2. Community extensions of the open-source library.
3. Agent-based simulation with non-exploitation agents.
4. Measurement development for exploitation detection (Section 15.7 roadmap).
5. Comparative case studies.
6. Machine learning estimation.
7. Cross-level dynamics.
8. Institutional evolution.
9. Polycentric design with justice constraints.
10. Compensation mechanism design and testing.
11. Strong vs. bounded non-exploitation comparison.
12. Intergenerational non-exploitation formalization.
13. Welfare function specification and testing.
14. Threshold determination methods.

18. Conclusion

This paper presents a multi-level extension of the General Governance Success Model (GSM), providing a computational systems framework for analysing how governance capabilities interact vertically and horizontally across nested scales, **subject to a bounded non-exploitation constraint that prohibits any jurisdiction from imposing uncompensated, non-trivial sacrifices on other jurisdictions for its own benefit.**

The extended model formalizes:

1. Multi-level governance structures.
2. Vertical and horizontal interactions, decomposed into cooperative and exploitative components.
3. A formal welfare function enabling constraint operationalization.
4. A bounded non-exploitation constraint admitting **threshold tolerance** and **vector-valued compensation**.
5. **A formal decomposition of the threshold parameter** $\tau_j = \tau(z_j, \omega)$ separating descriptive context from normative choice.
6. **Vector-valued compensation** $\text{Comp}_{i \rightarrow j} = (c_1, \dots, c_m)$ enabling testable compensation adequacy.

7. Information-flow fidelity, including transparency prerequisites for non-exploitation.
8. Distribution-sensitive aggregation (minimax, Gini-adjusted) that penalizes exploitation.
9. Recursive learning processes.
10. Constructive and destructive vertical misalignment.

The framework extends ideas from the Viable System Model, polycentric governance theory, multi-level governance theory, complex adaptive governance, network governance, and distributive justice theory, while providing a formal state-space representation suitable for empirical testing.

Important Note on the Empirical Example: The Denmark worked example and the comparative rankings are presented as **illustrative demonstrations of the GSM computational procedure**. While the raw indicator values are drawn from real data sources, the coefficients used to convert these indicators into GSM scores are **not empirically estimated**, and the rankings are **hypothetical**. Empirical validation remains an important direction for future research.

The model is released as an open-source Python package, `gsm-governance` (Wang, 2026b), which implements the state-space dynamics, the bounded non-exploitation constraint with counterfactual baseline and vector-valued compensation, the threshold decomposition, and the distribution-sensitive welfare and flourishing indices used in this paper. All numerical results reported in Section 16 can be reproduced with a single command.

The ultimate purpose of governance is not merely institutional preservation or ideological dominance, but the sustained improvement of human flourishing across generations and scales — **without sacrificing any jurisdiction for the benefit of another**. By focusing on governance functions rather than regime categories, by explicitly modelling governance across multiple levels, and by embedding a formal bounded non-exploitation constraint with threshold tolerance and vector-valued compensation, the extended GSM provides a foundation for comparative assessment, policy analysis, institutional design, and future empirical research that is both analytically rigorous and normatively defensible.

References

- Aarab, A., El Marzouki, A., Boubker, O., & El Moutaqi, B. (2025). Integrating AI in public governance: A systematic review. *Digital*, 5(4), 59. <https://doi.org/10.3390/digital5040059>
- Alon-Barkat, S., & Busuioc, M. (2024). Public administration meets artificial intelligence. *Journal of Behavioral Public Administration*, 7(1). <https://doi.org/10.30636/jbpa.71.261>
- Argyris, C., & Schön, D. A. (1978). *Organizational learning: A theory of action perspective*. Addison-Wesley.
- Ashby, W. R. (1956). *An introduction to cybernetics*. Chapman & Hall.
- Baldwin, E., Thiel, A., McGinnis, M., & Kellner, E. (2024). Empirical research on polycentric governance. *Policy Studies Journal*, 52(2), 319–348. <https://doi.org/10.1111/psj.12518>
- Beer, S. (1972). *Brain of the firm*. Allen Lane.
- Beer, S. (1979). *The heart of enterprise*. John Wiley & Sons.
- Checkland, P. (1981). *Systems thinking, systems practice*. John Wiley & Sons.

- Dahl, R. A. (1971). *Polyarchy: Participation and opposition*. Yale University Press.
- Espejo, R. (1994). A cybernetic method to study organizations. In R. Espejo & R. Harnden (Eds.), *The viable system model* (pp. 1–28). Wiley.
- Espejo, R. (2003). The viable system model: A briefing about organisational structure. *Syncho*, 1(1), 1–16.
- Evans, P. (1995). *Embedded autonomy*. Princeton University Press.
- Flood, R. L., & Jackson, M. C. (1991). *Creative problem solving: Total systems intervention*. John Wiley & Sons.
- Folke, C., Hahn, T., Olsson, P., & Norberg, J. (2005). Adaptive governance of social-ecological systems. *Annual Review of Environment and Resources*, 30, 441–473.
- Forrester, J. W. (1961). *Industrial dynamics*. MIT Press.
- Forrester, J. W. (1969). *Urban dynamics*. MIT Press.
- Fukuyama, F. (2013). What is governance? *Governance*, 26(3), 347–368.
- Funder, D. C., & Gardiner, G. (2024). Misgivings about measurement invariance. *European Journal of Personality*, 38(6). <https://doi.org/10.1177/08902070241228338>
- Grimmelikhuijsen, S., Jilke, S., Olsen, A. L., & Tummers, L. (2017). Behavioral public administration. *Public Administration Review*, 77(1), 45–56.
- Gunderson, L. H., & Holling, C. S. (Eds.). (2002). *Panarchy: Understanding transformations in human and natural systems*. Island Press.
- Hall, P. A., & Soskice, D. (Eds.). (2001). *Varieties of capitalism*. Oxford University Press.
- Holling, C. S. (1973). Resilience and stability of ecological systems. *Annual Review of Ecology and Systematics*, 4(1), 1–23.
- Holling, C. S. (2001). Understanding the complexity of economic, ecological, and social systems. *Ecosystems*, 4(5), 390–405.
- Hooghe, L., & Marks, G. (2001). *Multi-level governance and European integration*. Rowman & Littlefield.
- Hooghe, L., & Marks, G. (2003). Unraveling the central state, but how? *American Political Science Review*, 97(2), 233–243.
- Jackson, M. C. (2000). *Systems approaches to management*. Kluwer Academic.
- Jackson, M. C. (2003). *Systems thinking: Creative holism for managers*. Wiley.
- Janis, I. L. (1972). *Victims of groupthink*. Houghton Mifflin.
- Jilke, S., Bækgaard, M., Herd, P., & Moynihan, D. (2024). Short and sweet: Measuring experiences of administrative burden. *Journal of Behavioral Public Administration*, 7(1), 1–12. <https://doi.org/10.30636/jbpa.71.340>
- Kahneman, D. (2011). *Thinking, fast and slow*. Farrar, Straus and Giroux.
- Kahneman, D., & Tversky, A. (1979). Prospect theory. *Econometrica*, 47(2), 263–291.
- Klijn, E. H., & Koppenjan, J. (2016). *Governance networks in the public sector*. Routledge.
- Kusano, K., Napier, J. L., & Jost, J. T. (2026). The mismeasure of culture. *Personality and Social Psychology Bulletin*.
- Levin, S., Xepapadeas, T., Crépin, A. S., Norberg, J., de Zeeuw, A., Folke, C., ... & Walker, B. (2013). Social-ecological systems as complex adaptive systems. *Environment and Development Economics*, 18(2), 111–132.
- Mansbridge, J. (2003). Rethinking representation. *American Political Science Review*, 97(4), 515–528.

- Midgley, G. (2000). *Systemic intervention: Philosophy, methodology, and practice*. Kluwer Academic.
- Miller, D. (2013). *Justice for earthlings: Essays in political philosophy*. Cambridge University Press.
- Morgan, M., Webster, A., Piccarello, M., Jones, K., Chermak, J., McCarthy, L., & Srinivasan, J. (2023). Adaptive governance strategies. *Frontiers in Climate*, 5. <https://doi.org/10.3389/fclim.2023.1062320>
- Morrison, T. H., Bodin, Ö., Cumming, G. S., Lubell, M., Seppelt, R., Seppelt, T., & Weible, C. M. (2023). Building blocks of polycentric governance. *Policy Studies Journal*, 51(3), 475–499.
- Nussbaum, M. C. (2011). *Creating capabilities: The human development approach*. Harvard University Press.
- OECD. (2022). *How's life? 2022: Measuring well-being*. OECD Publishing.
- Ostrom, E. (1990). *Governing the commons: The evolution of institutions for collective action*. Cambridge University Press.
- Ostrom, E. (2010). Polycentric systems for coping with collective action and global environmental change. *Global Environmental Change*, 20(4), 550–557.
- Pearl, J. (2009). *Causality* (2nd ed.). Cambridge University Press.
- Peña Pazos, G. L., Fernández Miranda, M., et al. (2026). Anticipatory governance and artificial intelligence. *Administrative Sciences*, 16(7), 326. <https://doi.org/10.3390/admsci16070326>
- Provan, K. G., & Kenis, P. (2008). Modes of network governance. *Journal of Public Administration Research and Theory*, 18(2), 229–252.
- Putnam, R. D. (1993). *Making democracy work: Civic traditions in modern Italy*. Princeton University Press.
- Putnick, D. L., & Bornstein, M. H. (2016). Measurement invariance conventions and reporting. *Developmental Review*, 41, 71–90. <https://doi.org/10.1016/j.dr.2016.06.004>
- Rawls, J. (1971). *A theory of justice*. Harvard University Press.
- Rhodes, R. A. W. (1996). The new governance. *Political Studies*, 44(4), 652–667.
- Rodrik, D. (2000). Institutions for high-quality growth. *Studies in Comparative International Development*, 35(3), 3–31.
- Rothstein, B. (2011). *The quality of government*. University of Chicago Press.
- Russell, S., & Norvig, P. (2020). *Artificial intelligence: A modern approach* (4th ed.). Pearson.
- Saikia, P., & Jiménez, A. (2023). Governance attributes for building water resilience. *Water International*. <https://doi.org/10.1080/02508060.2023.2274162>
- Sangiovanni, A. (2007). Global justice, reciprocity, and the state. *Philosophy & Public Affairs*, 35(1), 3–39.
- Schwaninger, M. (2004). Methodologies in conflict. *Systems Research and Behavioral Science*, 21(5), 543–556.
- Schwaninger, M. (2006). *Intelligent organizations*. Springer.
- Sen, A. (1999). *Development as freedom*. Oxford University Press.
- Senge, P. M. (1990). *The fifth discipline*. Doubleday.
- Simon, H. A. (1957). *Models of man: Social and rational*. John Wiley & Sons.
- Skocpol, T. (1985). Bringing the state back in. In P. B. Evans, D. Rueschemeyer, & T. Skocpol (Eds.), *Bringing the state back in* (pp. 3–37). Cambridge University Press.
- Sterman, J. D. (2000). *Business dynamics*. McGraw-Hill.

Stiglitz, J. E., Sen, A., & Fitoussi, J.-P. (2009). *Report by the Commission on the Measurement of Economic Performance and Social Progress*.

Sutton, R. S., & Barto, A. G. (2018). *Reinforcement learning* (2nd ed.). MIT Press.

Tajfel, H., & Turner, J. C. (1979). An integrative theory of intergroup conflict. In W. G. Austin & S. Worchel (Eds.), *The social psychology of intergroup relations* (pp. 33–47). Brooks/Cole.

Ulrich, W. (1983). *Critical heuristics of social planning*. Haupt.

United Nations Development Programme. (2022). *Human Development Report 2021/2022*.

Vandenberg, R. J., & Lance, C. E. (2000). A review and synthesis of the measurement invariance literature. *Organizational Research Methods*, 3(1), 4–70. <https://doi.org/10.1177/109442810031002>

Vickers, G. (1965). *The art of judgment*. Chapman & Hall.

von Foerster, H. (1974). *Cybernetics of cybernetics*. University of Illinois.

Wang, H. (2026a). A computational systems model of governance. *Preprint*, [Preprints.org](https://doi.org/10.20944/preprints202609.0233.v1). <https://doi.org/10.20944/preprints202609.0233.v1>

Wang, H. (2026b). *gsm-governance: A Python library for the General Governance Success Model with bounded non-exploitation* (Version 0.2.0) [Computer software]. <https://github.com/hongxueharriswang/gsm-governance>

Wiener, N. (1948). *Cybernetics*. MIT Press.

Young, I. M. (2011). *Responsibility for justice*. Oxford University Press.

Appendix A: Multi-Level Notation Table

Symbol	Description	Dimensions
$\mathcal{L}$	Set of governance levels	$\{1, 2, \dots, L\}$
$\mathbf{g}_{i,\ell,t}$	Governance state at level ℓ	$\mathbb{R}^5$
$\mathbf{f}_{i,\ell,t}$	Flourishing state at level ℓ	$\mathbb{R}^4$
$\Delta_{g,\ell}$	Governance depreciation at level ℓ	$\mathbb{R}^{5 \times 5}$
$\Delta_{f,\ell}$	Flourishing depreciation at level ℓ	$\mathbb{R}^{4 \times 4}$
$\mathbf{T}_{\ell+1 \rightarrow \ell}$	Top-down influence matrix	$\mathbb{R}^{5 \times 5}$
$\mathbf{T}_{\ell+1 \rightarrow \ell}^{\text{coop}}$	Cooperative top-down influence	$\mathbb{R}^{5 \times 5}$
$\mathbf{T}_{\ell+1 \rightarrow \ell}^{\text{exploit}}$	Exploitative top-down influence	$\mathbb{R}^{5 \times 5}$
$\mathbf{B}_{\ell-1 \rightarrow \ell}$	Bottom-up influence matrix	$\mathbb{R}^{5 \times 5}$
$\mathbf{H}_{\ell}$	Horizontal coordination matrix	$\mathbb{R}^{5 \times 5}$
$\mathbf{M}_{i,\ell,t}$	Information-flow matrix at level ℓ	$[0, 1]^{5L \times 5L}$

Symbol	Description	Dimensions
$F_{\ell \rightarrow k}$	Information fidelity matrix	$[0, 1]^{5 \times 5}$
$W_{D,\ell}$	Misalignment weighting matrix	$\mathbb{R}^{5 \times 5}$
W_{ℓ}^{vert}	Vertical misalignment weighting matrix	$\mathbb{R}^{5 \times 5}$
ω_{ℓ}	Governance aggregation weight	$\mathbb{R}_+$
v_{ℓ}	Flourishing aggregation weight	$\mathbb{R}_+$
$\lambda_1, \lambda_2, \lambda_3$	Misalignment benefit, cost, exploitation penalty parameters	$\mathbb{R}_+$
$\mathcal{E}_{i,\ell,t}$	Exploitation detection indicator	$\{0, 1\}$
$\mathcal{U}_{i,\ell,t}^{\text{NE}}$	Bounded non-exploitation feasible action set	Subset of $\mathbb{R}^5$
$\mathcal{W}_j$	Jurisdiction j 's welfare measure	$\mathbb{R}$
θ_g, θ_f	Welfare weights	$\mathbb{R}_+$
$\mathbf{z}_j$	Contextual variables for jurisdiction j	$\mathbb{R}^k$
$\boldsymbol{\omega}$	Normative parameters for threshold determination	$\mathbb{R}^m$
$\tau_j = \tau(\mathbf{z}_j, \boldsymbol{\omega})$	Threshold for acceptable sacrifice	$\mathbb{R}_+$
$\text{Comp}_{i \rightarrow j}$	Vector of compensation transfers from i to j	$\mathbb{R}_+^m$
$\boldsymbol{\lambda}_j$	Recipient-specific compensation valuation vector	$\mathbb{R}_+^m$
$E^{\text{env}}, E^{\text{fiscal}}, E^{\text{reg}}, E^{\text{inst}}, E^{\text{temp}}$	Exploitation index components	$\mathbb{R}_+$
$w_1, \dots, w_5$	Exploitation index component weights	$\mathbb{R}_+$
ρ	Inequality aversion parameter	$\mathbb{R}_+$
$D_{i,t}^{\text{vert}}$	Vertical misalignment measure	$\mathbb{R}_+$
$RV_{i,t}$	Requisite variety ratio	$\mathbb{R}_+$
$\mathcal{A}_{i,\ell,t}$	Level-specific adaptability	$\mathbb{R}^{5 \times 5}$

Appendix B: Data Sources and Indicators

GSM Dimension	Proxy Indicator	Data Source	URL
Accountability	V-Dem Liberal Democracy Index	V-Dem Institute	https://www.v-dem.net

GSM Dimension	Proxy Indicator	Data Source	URL
Competence	Government Effectiveness Index	World Bank WGI	https://info.worldbank.org/governance/wgi/
Cohesion	WJP Rule of Law Index	World Justice Project	https://worldjusticeproject.org
Continuity	Human Development Index	UNDP	https://hdr.undp.org
Learning	Government Service Satisfaction	Our World in Data	https://ourworldindata.org